

Site experience # 5.0

Introduction.....	1
Reflective - What I learnt ?.....	1
On the tools experience.....	1
Testing end-to-end.....	1
Figure of eight test.....	2

Introduction

On **Wednesday 26th November 2025** I gained valuable practical experience across domestic properties in Leicestershire.

Reflective - What I learnt ?

On the tools experience

I learnt that when you have a single 2.5 mm² use short-nose pliers to bend back the copper wire in order to create more copper surface area for the terminal to grip. This improves the connection.

Testing end-to-end

We visited a few different residences on this day however, I wanted to focus on the last site visit. For a basic overview, the homeowner built an extension to enlarge the kitchen. As part of the extension there were some sockets that had to be installed in a ring final circuit.

The first test was an end to end test.

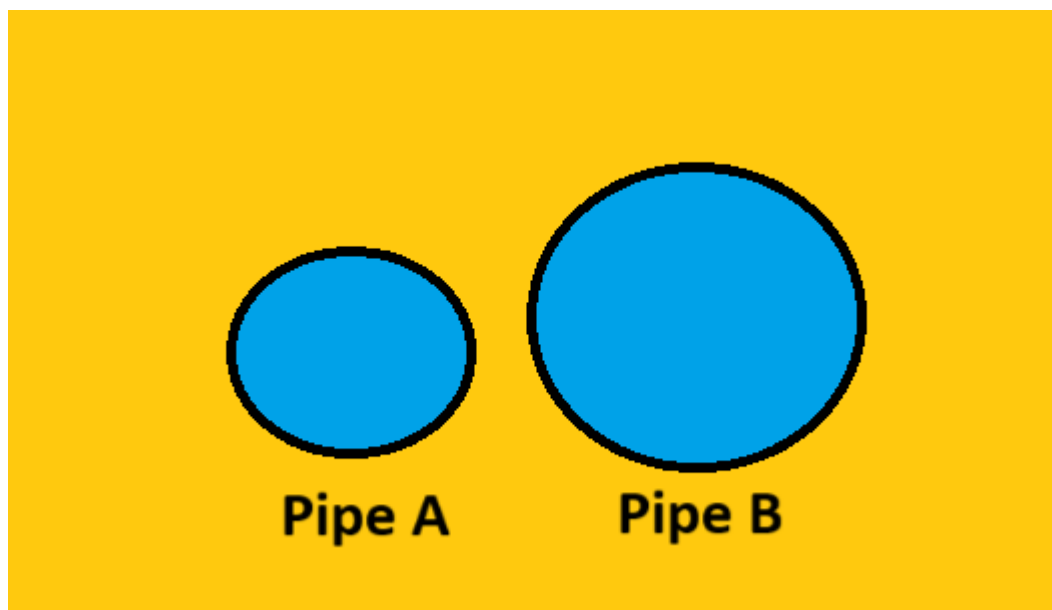
A end to end is when you test:

Brown live from **purple cable one** with the **brown live** from **purple cable two**;
Blue neutral from **purple cable one** with **blue neutral** from **purple cable two**;
Green and **yellow** CPC from **purple cable one** with **Green** and **yellow** CPC from **purple cable two**.

An end-to-end test for example with the brown live wires. Imagine that you stretch out the brown live wire as far as possible. With that test you want to confirm that electrons flow from

Thursday 27th November 2025
IWA

one end of the brown live wire to the other end. However, in reality we have sockets and fuse connection units and other appliances that increase resistance and cause voltage drop. For an end-to-end test we expect the ohms (resistance) of the CPC to be higher than both the brown line and blue neutral wires. The reason for this is because the CPC is smaller than both the brown line and blue neutral wire. This means there is less space. A good analogy is if we have two water pipes. Pipe A and pipe B.



The water pressure in **pipe A** will be higher than **pipe B**. This is because in **pipe A**, there is less room/space for the water to flow. Electrons behave in the same manner as water particles. The resistance (measured in ohms) of a CPC wire will be higher compared to the blue neutral and brown live wire. Because, the latter two wires have a greater cross-sectional area which means a lower resistance value.

Figure of eight test

We also conducted a figure of eight tests.

In simple terms, a figure of eight test is when you test:

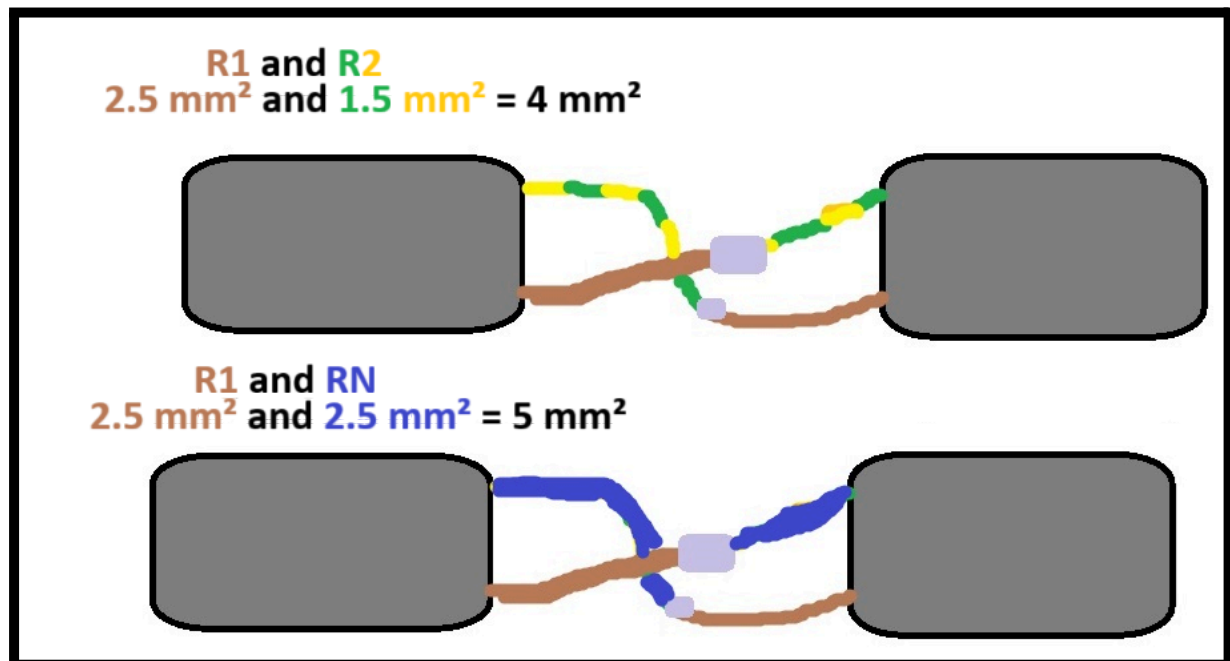
The **brown live wire** from **cable one**. With the **cpc** from **cable two**.

The **brown live wire** from **cable two** with the **cpc** from **cable one**.

Thursday 27th November 2025
IWA

The **brown live wire** from **cab**le one with the **blue neutral wire** from **cab**le two.
The **brown live wire** from **cab**le two with the **blue neutral wire** from **cab**le one.

Depiction of the figure of eight test



The figure of eight does not have to be performed at the fuse board and can be arranged at a socket.

Once you have organised for a figure of eight test use a tester plug and go around each and every socket on the ring circuit and note down the highest reading.

One area which was a cause of confusion was I thought that the ohms resistance for **R1** and **RN** would be higher than the ohms resistance for **R1** and **R2**. However, this is incorrect. **Because, the resistance is always higher in wires which have a smaller cross-sectional area.** As you can see in the diagram "Depiction of the figure of eight test" a **R1** and **R2** has a small cross sectional area. However, as in the water pipe analogy. The smaller the size and higher the resistance.

The **higher** the combined cross sectional area - the **lower** the resistance
The **smaller** the combined cross-sectional area - the **higher** the resistance